

Yaseen Mohamed

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Education

University of Waterloo | BSc, Mechatronics Engineering **Avg: 88.6%** | Excellent Standing Grad 2028

- **Relevant Coursework:** Embedded Systems, OS & RTOS, Sensors & Instrumentation, Signals & Systems, Data Structures & Algorithms, Actuators & Power Electronics.

Skills

- **Embedded/Firmware:** C/C++, ARM Cortex-M, RTOS, DMA, ADC, timers/interrupts, PWM, Linux (Bash/CLI).
- **Interfaces/Debug:** CAN/CAN FD, I2C, SPI, UART, Ethernet, Wi-Fi, Bluetooth; JTAG/SWD, ST-Link, OpenOCD, GDB.
- **Platforms:** STM32 (HAL/LL), Arduino, ESP32, Raspberry Pi.
- **Software/Tools:** Python (Pytest, Pandas, NumPy), MATLAB, SQL, Git, Azure DevOps, Jira, Robot Framework.
- **Hardware/CAD:** PCB design (Altium), soldering, oscilloscope, DMM, bench power supply, SolidWorks, Inventor, AutoCAD, 3D printing.

Experience

System Integration Engineer

May – Aug 2026

Ford Motor Company

- Tested software-update failure scenarios on a **cloud-connected vehicle integration bench**, configuring ECU DIDs and cloud connectivity, flashing multiple software versions, testing OTA updates, and analyzing CAN and Ethernet logs.
- Traced **gateway ECU C++ code** and system logs to identify potential OTA failure paths, and created code-navigation and investigation guides that reduced investigation time from **1–2 hours to 5–15 minutes**.
- Built **4 Python investigation and automation tools** using Tkinter, Pandas, and Plotly to process datasets containing 7M+ rows, reducing hands-on investigation time by **up to 90%**.
- Investigated large-scale vehicle software-update issues using vehicle logs, Google BigQuery, SQL, and Python across **1M–80M-row datasets**, validating edge cases and providing technical evidence for Jira investigations.
- Developed and optimized BigQuery queries using CTEs, window functions, partition filters, JSON parsing, and reusable tables, reducing runtime by **93%** and recurring data-retrieval time from **2 hours to 10–15 minutes**.

Embedded Software Engineering Intern

Sep – Dec 2025

Aversan Inc.

- Debugged and extended **STM32H7 embedded C firmware** for a safety-critical aerospace device under **DO-178 processes**, performing bench validation and delivering verified firmware changes through the team's development workflow.
- Built **5 PEP 8-compliant Python tools** using UART to parse and log test data and automate firmware debugging and verification, identifying dozens of firmware and test defects and cutting test time by **70%**.
- Built a Linux Python GUI for **JTAG boundary-scan (OpenOCD)** to detect soldering defects, short circuits, and connectivity issues, and run **bit-banged I2C** tests without MCU firmware, catching manufacturing faults early and reducing rework.
- Implemented GPIO bit-banged I2C sensor and peripheral interfacing on a custom STM32H7 board, and used DMA, HAL, custom timers, and **interrupt-driven UART RX** to automate testing and status reporting, reducing component test time by **90%**.
- Created automated I/O tests in **Robot Framework** (via RIDE) on Linux-based Automated Test Equipment (ATE) to verify set/read behavior and generate reports supporting debugging, FAA certification, production, and maintenance.

Engineering Research Assistant

Jan – Apr 2025

University of Waterloo School of Optometry and Vision Science

- Built a pressure-sensing footstep detection system using an Arduino, 48-channel MUX, custom PCB, and Raspberry Pi, then developed a **decay-compensated C++/Python acquisition pipeline** that reduced the sensor sampling interval from **3 s to 3 ms**.
- Built a Python GUI with a **grouping algorithm** to preprocess and visualize foot-pressure data, and designed SolidWorks/FDM equipment upgrades that reduced setup time by **80%**.

Projects

2-Axis STM32 Machine Controller | [GitHub](#) · [Demos](#) | C, STM32, SPI, ADC, UART, Python May – Jul 2026

- Developed **STM32 firmware in C** for independent dual-axis stepper control via SPI drivers, with dual ADC-based speed adjustment and UART commands from a Tkinter/PySerial GUI.
- Implemented **interrupt-driven limit-switch safety logic** with wiring-fault detection and direction-aware motion blocking, and prevented an **ADC/EXTI race condition** using persistent per-axis direction and stop states.

Quadcopter Prototype | [GitHub](#) · [Demo](#) | C++, Arduino, IMU, Timers, PID Control Jun – Aug 2020

- Built an Arduino quadcopter prototype with **PID stabilization** in C++, using **IMU gyroscope feedback** and a custom PCB to implement and tune a real-time closed-loop attitude controller.